Adaptable Vehicle Detection and Speed Estimation for Changeable Urban Traffic With Anisotropic Magnetoresistive Sensors

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***Abstract*— This paper presents an adaptable roadside vehicle detection and speed estimation system for various traffic conditions on urban roads based on tri-axial anisotropic magnetoresistive sensors and wireless sensor network. The system consists of one master node and two sensor nodes, which are placed along the roadsides and can measure the earth's local magnetic field disturbance caused by passing vehicles. A dynamic threshold detection algorithm is proposed for vehicle detection, especially considering the actual variable traffic condition. The vehicle speed is estimated on the basis of the maximum values and the cross correlation of effective parts extracted from two sensor signals. We have tested the vehicle information at several roads under different traffic conditions. Validation study has revealed a high detection accuracy of 97.92% when using a dynamic threshold compared with 92.3% when using a fixed threshold. And, the average accuracy of speed estimation can reach up to 97.11% on the roads. The proposed algorithm has a significant increase in accuracy, reliability, and practicability compared with the fixed threshold algorithm.**

***Index Terms*— Anisotropic magnetoresistive sensor (AMR), vehicle detection, speed estimation, changeable urban traffic.**

I. Introduction

VEHICLE detection and speed measurement technologies are used to collect traffic information in the Intelligent Transportation System (ITS). Nowadays, vehicle detection systems commonly use inductive loop detector [1], video camera [2] and radar detector [3], etc. The installation and maintenance of inductive loop requires interrupting transportation, which will lead to a high cost. The video camera is greatly affected by the environment, such as light intensity. The radar detector has poor adaptability to the weather and can be easily influenced by temperature and air flow. All of them are expensive for a large-scale deployment. Recently, the magnetic sensor has attracted great attention due to its low cost and low power consumption. It can be used with wireless sensor network for vehicle detection, which is reliable, easily installed and can reach high accuracy [4].

Researchers have proposed many vehicle detection methods with magnetic sensors. One approach is using a cross-correlation between the measured signal and the reference signal to detect vehicle [5]. However, finding the proper reference signal is not easy, which will cause high power computation. In contrast, the threshold-based methods are simple and easy-to-use. Balid *et al.* [6] updates the localized magnetic field reference components and analyze various vehicles to decrease the detection error. This algorithm can detect various vehicles with different speeds successfully. In our previous work, we proposed a fixed threshold state machine algorithm based on signal variance [7]. It can suppress the signal drift caused by changed temperature and the detection accuracy can reach up to 99.05%. However, the urban traffic is complex and changeable, and the fixed thresholds cannot adapt to different traffic conditions. Therefore, adjusting thresholds according to the changeable traffic condition simply and automatically will be a new direction for further research.

To estimate the vehicle speeds, a single sensor [8] or two longitudinally spaced sensors [9] are generally used. A speed estimation algorithm based on the calculation of cross correlation was proposed in [10], and the time delay can be obtained based on the maximum value of the calculation results. Saber Taghvaeeyan et al. used the fast Fourier transform (FFT) to calculate the cross correlation, which can reduce computation and greatly improve accuracy to 97.5% [11]. However, the algorithms mentioned above only considered the case of a single vehicle passing by the sensors. In urban traffic, the vehicles always crawl bumper to bumper, so we need to extract the effective signal of each vehicle from sensor signal. The conventional vehicle signal extraction method is based on the vehicle entering and leaving points [7], [12]. Slight changes of the detection points will result in a great change in the vehicle speed, especially with a low sampling frequency in wireless sensor network. Hence decreasing the impact of these vehicle detection points will increase the speed estimation accuracy.

In this paper, we propose an adaptable vehicle detection and speed estimation system, which can be used in the changeable and complex urban traffic. The system is based on triaxial anisotropic magnetoresistive (AMR) sensors and wireless sensor network, and it can be placed along the roadside without any influence on the traffic. A dynamic threshold detection algorithm is proposed to detect vehicles, and the thresholds can be dynamically adjusted according to the traffic condition. The maximum values of the two sensors and the cross correlation between them are analyzed to extract vehicle signals and estimate vehicle speed. The experimental results have shown that the proposed algorithms can fit the urban traffic with high accuracy.

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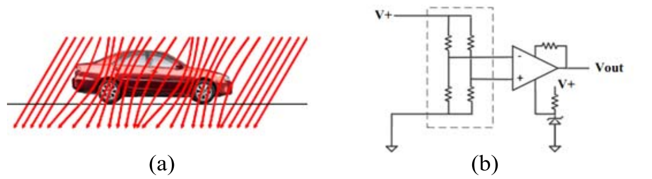

Fig. 1. Principle of vehicle detection based on magnetic sensor: (a) the disturbance of geomagnetic field by a moving vehicle [13] and (b) AMR sensor bridge.

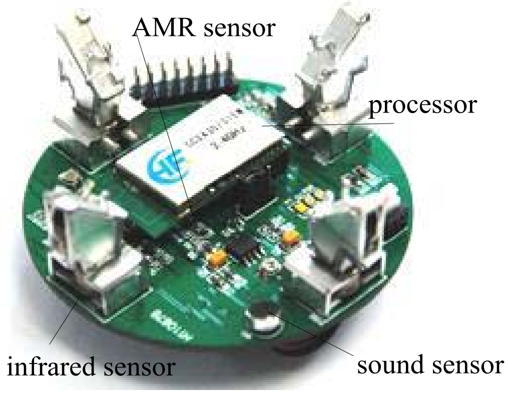

Fig. 2. The sensor node in the sensing system.

This paper is organized as follows. Section II introduces the configuration of the sensing system. Section III describes the algorithm for vehicle detecting considering the actual traffic condition. The speed estimation method is discussed in section IV. Section V shows the experimental results of vehicle detection and vehicle speed estimation. Finally, some conclusions are contained in section VI.

II. Sensing System Configuration

The geomagnetic field can be disturbed by the ferromagnetic material in vehicles. The magnetic sensor is used to detect the variation of geomagnetic field and can output a voltage value by a Wheatstone bridge. Figure 1 shows the principle of vehicle detection based on magnetic sensor.

The sensor system and the sensor setup used in this paper are the same as that in [7]. We use the Honeywell HMC5883L triaxial anisotropic magnetoresistive sensor in this system due to its high level of integration and low cost. The sensor has a build-in 12 bits analogue-to-digital converter (ADC) and achieves $0.5\mu T$ resolution in $\pm 800.0\mu T$ Fields. The average current drain of the sensor is only $100.0\mu A$ and as low as $2.0\mu A$ in standby mode. Figure 2 shows one of the two sensor nodes in the system, which contains a CC2430 processor and a magnetic sensor. There are also four infrared sensors and a sound sensor in the sensor node for other measurement tasks. All the nodes are battery-powered with small size, and they can be easily installed along the roadside.

Figure 3 shows the deployment of the system consisting of one master node and two sensor nodes. The sensor nodes and master node are connected wirelessly, and the measured data can be transferred to a laptop through a RS232 serial port. The master node works as the clock source and provides

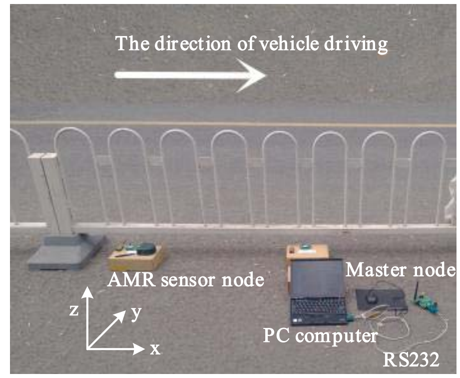

Fig. 3. The configuration of the sensor system.

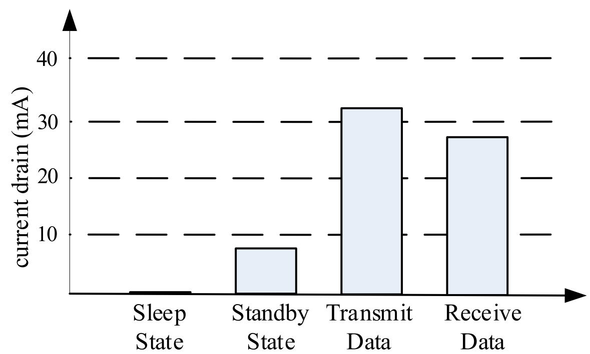

Fig. 4. Current drain of the sensor node.

time synchronization for all the nodes. The master node is in the working state all the time, and the sensor nodes work in sleep and wake-up state. Through polling mechanism, the master node can receive the data from the two sensor nodes successively. Figure 4 shows the current drain of the sensor nodes working in different states. The current drain is much higher when the sensor node is transmitting and receiving data. To reduce the power consumption, vehicle detection and speed estimation algorithm with a low sampling frequency is concerned. The sampling frequency of each sensor node is set as 25 Hz and defined as f_s in this paper. Sensor node 1 is used to detect vehicles and the detection algorithm is described in Section III. Sensor node 2 is used to estimate the vehicle speed along with sensor node 1 and the method is discussed in Section IV. Considering the similarity of the two sensor nodes and the sampling frequency, the distance between the two sensor nodes is experimentally set as 3 m. The x-axis of magnetic sensor is parallel to the direction of the moving vehicles, the y-axis is perpendicular to the direction of the moving vehicles, and the z-axis is perpendicular to the ground.

III. Vehicle Detection

A. Multi-State Machine

The fluctuation of geomagnetic field is relatively weak and can be disturbed easily by the surrounding electromagnetic field. Figure 5 (a) shows the signals measured along the x-, y-, and z-axis respectively for a two-box vehicle passing in the near road. The horizontal axis represents the sampling points of the magnetic sensor, while the vertical axis represents

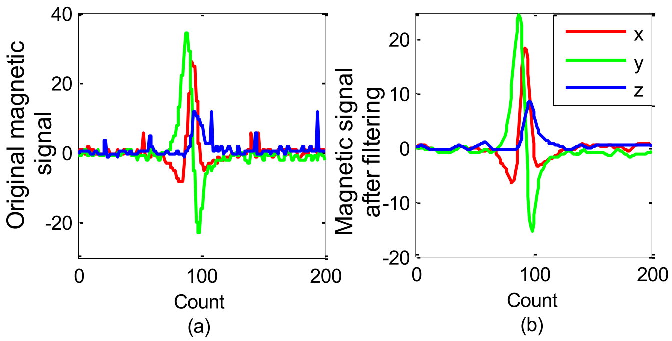

Fig. 5. Magnetic signals of a two-box vehicle: (a) original signals and (b) signals after filtering.

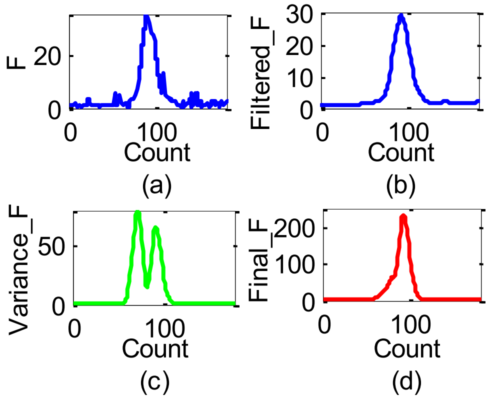

Fig. 6. Signals of a two-box vehicle after processing: (a) F and (b) F after filtering and (c) the variance of F and (d) the final signal after processing.

the intensity of magnetic field. A median filter and a moving average filter are used and the filtered signals are shown in Fig. 5(b).

We use the data preprocessing algorithm in our previous work [7]. There will be a drift in background signal when the temperature changes [14], and this algorithm can not only reduce the influence of the signal drift, but also retain the characteristics of the magnetic field intensity signal.

F is defined as the magnetic field intensity here:

$$F = \sqrt{X^2 + Y^2 + Z^2}, \tag{1}$$

where X, Y, and Z are the magnetic readings of x-, y-, and z-axis of magnetic sensor respectively. The final signal used to detect vehicles is called $Final\_F$, which is given by

$$Final\_F = Filtered\_F \times \sqrt{Variance\_F}, \tag{2}$$

where $Filtered\_F$ is the signal of F after filtering and $Variance\_F$ is the variance of $Filtered\_F$. The signals of the two-box vehicle mentioned above are shown in Fig. 6.

Vehicles on the adjacent road can also affect the AMR sensor output, and an experiment was performed to analyze the influence. The system layout is illustrated in Fig. 7. The vehicle passed by the sensors on the adjacent road, and its signals are shown in Fig. 8. The amplitudes of signals are much smaller than the signals shown in Fig. 6. As a result, the influence caused by the vehicles on adjacent road can be neglected compared with those on the near road.

In our previous work, we used a fixed threshold state machine algorithm based on signal variance. We consider

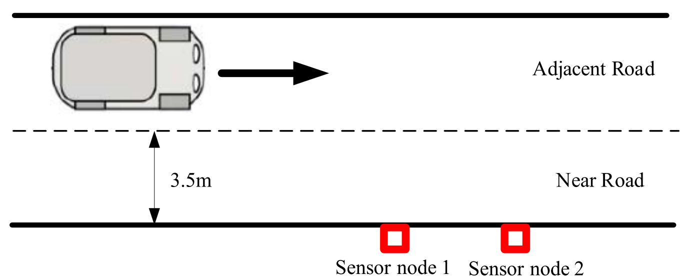

Fig. 7. Sensor installation diagram.

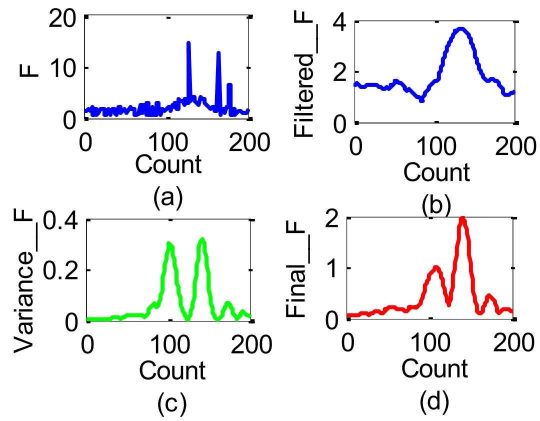

Fig. 8. Signals of a two-box vehicle on the adjacent road: (a) F and (b) F after filtering and (c) the variance of F and (d) the final signal after processing.

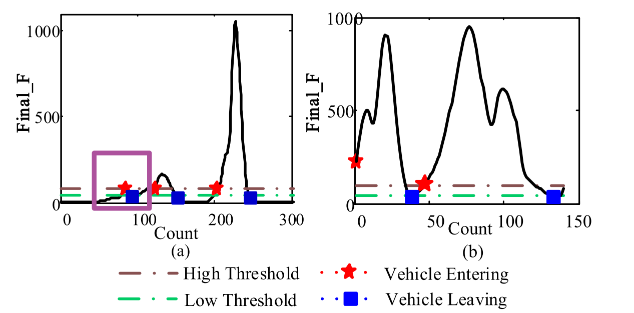

Fig. 9. The misjudgment results using our previous algorithm: (a) a jamming signal misjudged as one vehicle and (b) one vehicle misjudged as two.

the vehicle entering and leaving the sensor monitoring area when the value of final signal is just higher than the High Threshold and just lower than the Low Threshold, respectively. This algorithm, however, may lead to wrong judgment on vehicle detection. Fig. 9 (a) shows that a jamming signal is regarded as a vehicle signal and the interference is caused by a wheelbarrow passing by the sensor. Fig. 9 (b) shows that a low-speed vehicle (about 1.7 m/s) is mistaken for two vehicles.

Referring to the multi-state machine proposed in [6] and [15], we add two intermediate states on the basis of the previous one. It still uses the deviations of $Final\_F$ from High Threshold and Low Threshold to detect vehicles. The schematic of this algorithm is presented in Fig. 10 and the input is defined as SI, which is given by

$$SI = \begin{cases} 0 & Final\_F(n) \leq Low\ Threshold \\ 2 & Final\_F(n) \geq High\ Threshold \\ 1 & otherwise, \end{cases} \tag{3}$$

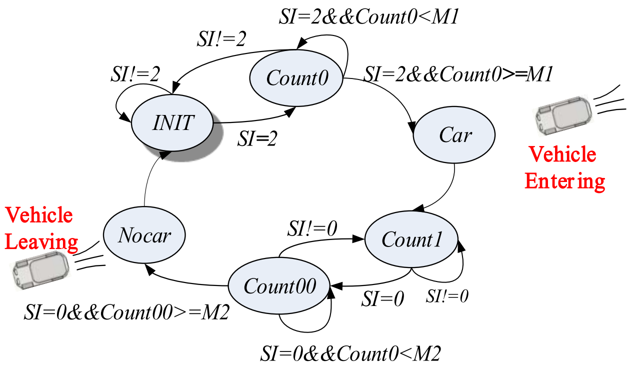

Fig. 10. State machine used in the sensor system for vehicle detection.

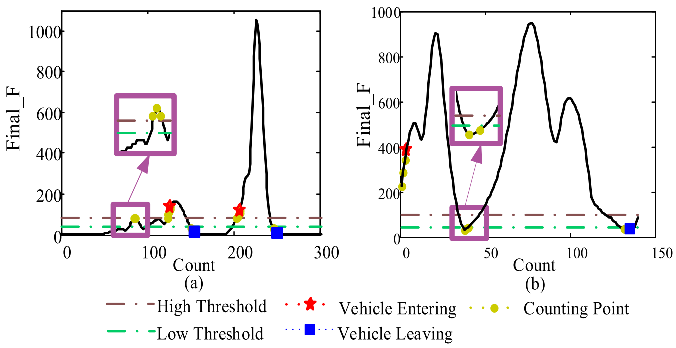

Fig. 11. Vehicle detection results using our improved algorithm: (a) a jamming signal judgment and (b) one vehicle judgment.

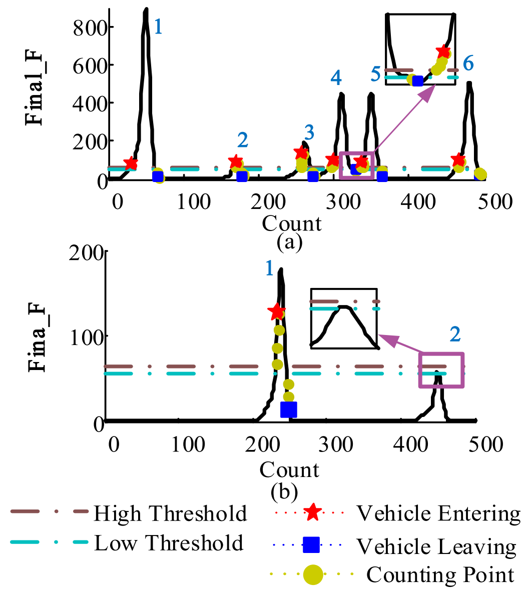

Fig. 12. Vehicle detection results based on the fixed thresholds algorithm: (a) $NUM = 6$ and (b) $NUM = 2$.

where $Final\_F(n)$ represents the corresponding value of $Final\_F$ of the n*th* sampling point. $INIT$ counts for the initial state, and all parameters are set to initial values. $Count0$ is the intermediate state which represents counter for vehicles entering the sensor monitoring area. When the vehicle signal is higher than the High Threshold, the counter starts to count the number, and the counting results are compared with counting threshold set as $M1$. If the signal becomes lower than the High Threshold when the counting result is still smaller than $M1$, the counter will be cleared to 0. $Count00$ represents counter for vehicle leaving the sensor monitoring area, and the counting result is compared with $M2$. By adding the two intermediate states, we expect to recognize the interference and improve the stability of this algorithm. $Count1$, in addition, is used to record the lasting time for vehicles staying in the sensor monitoring area. Finally, Car and $Nocar$ are the outputs of this state machine.

Figure 11 shows the detection results of our improved algorithm for small disturbances and very low speed vehicle, and the detection accuracy is clearly improved.

B. Dynamic Threshold Detection Algorithm

The accuracy of vehicle detection is closely related to the values of High and Low Thresholds, and how to select the proper thresholds is discussed in this paper. In urban traffic, the traffic condition is changeable, so the following experiments were performed in order to find the relationship between the two thresholds and the traffic condition. Note that the target used to describe the traffic condition is called NUM, which refers to the total number of passing vehicles in about 20 seconds (the time required for sampling 500 points).

The sensing system was placed along Xueyuan Road in Haidian District, Beijing. The vehicles driving on the road includes two-box vehicle, saloon, sports utility vehicle (SUV) and van. The speed ranges from 4-14m/s and the number of vehicles is bigger in the rush hour. To analyze the detection accuracy under different traffic condition, we test two different cases: $NUM = 6$ and $NUM = 2$, and these 8 vehicles are all two-box vehicles. Figure 12 shows the detection results of the two signals based on the same thresholds. For the case $NUM = 6$ and$NUM = 2$, their optimal thresholds values have no intersection. In Figure 12, the chosen thresholds are optimal for the case $NUM = 6$, while being too high for the case $NUM = 2$. As for our previous work in [7], the traffic condition is limited during the test, which makes the optimal thresholds can be found easily, and the vehicle detection can achieve high accuracy. However, the fixed threshold algorithm is not applicable for variable urban traffic, which means poor adaptability of the algorithm. A dynamic threshold detection algorithm is required.

Our previous work presented an adaptive threshold detection algorithm, which detects vehicles with constantly-changing thresholds. The thresholds will change as long as there is a slightly change of the sensor signal, which results in a poor anti-jamming ability of the algorithm. The adaptive thresholds change depend on the variance of the history signal. When the low speed vehicles pass by the sensor, the adaptive threshold increases due to the influence of the front vehicle, so the thresholds keep rising correspondingly. Sometimes, the threshold adjustments are too high for the purpose of proper vehicle detection [7]. Hence the detection result, especially when the vehicles crawl bumper to bumper with low-speed in urban traffic, is not satisfactory.

A dynamic threshold multi-state machine algorithm is proposed to improve the capacity of resisting interference and

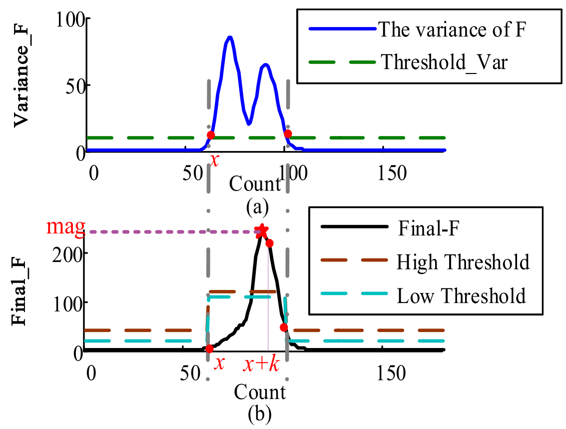

Fig. 13. Determination of the High and Low Thresholds: (a) the value of x and (b) the values of mag and the two thresholds.

avoid over frequent threshold adjustments. $Variance\_F$ is used as a preliminary judgment signal and $Final\_F$ is used to determine the values of High and Low (Fig.13). Firstly, we set an initial threshold for $Variance\_F$, which can be defined as $Threshold\_Var$. There may be a vehicle entering when $Varianced\_F$ is just higher than it, and the sampling point can be defined as x. Then we can extract a segment of the $Final\_F$ based on the value of x

$$seg = \{Final\_F(x), Final\_F(x+1), \ldots, Final\_F(x+k)\}, \tag{4}$$

where k is experimentally set to 30.

Define

$$mag = \max(seg). \tag{5}$$

Notice that mag happens to get the maximum of $Final\_F$, but it is not always the case. If the speed of vehicle is too slow, mag occurs before the maximum value. Now we can get the High and Low Thresholds by

$$\begin{cases} High\ Threshold = mag/2; \\ Low\ Threshold = l^{*}mag/2; \quad 0 < l < 1, \end{cases} \tag{6}$$

where l is experimentally set to 0.9. Then if the value of $Variance\_F$ is sustainably higher than $Threshold\_Var$, the High and Low Thresholds remain unchanged. When it is lower than $Threshold\_Var$, the High and Low Thresholds are resumed to the initial value. By this way, the two thresholds are dynamically adjusted according to the change of sensor signal when the vehicle is entering, and they will not be too high to detect vehicles.

$M1$ and $M2$, in addition, also influence the detection result. The recognizition of the noise affects the selection of $M1$. During the experiments, the duration of the noise is always less than 150 ms. As the sampling period is 40 ms, considering the High Threshold, $M1$ is set as 4 (160ms) in the experiments. When the vehicle speed is small enough, $Final\_F$ will fluctuate and the signal may be lower than the Low Threshold during the time when the vehicle is passing by the sensor. Increasing $M2$ can prevent the misdetection. We tested about 3000 vehicles, and almost all of the vehicles

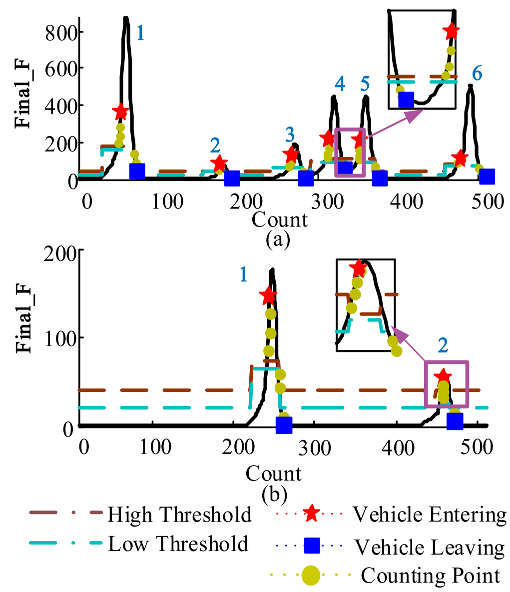

Fig. 14. Vehicle detection results using our improved algorithm: (a) $NUM = 6$ and (b) $NUM = 2$.

arrive beyond 200ms. Considering the Low threshold, $M2$ is set as 3 (120ms) in the experiments.

Figure 14 shows the detection results using the improved algorithm. After adjusting the thresholds, the vehicles can be detected under different traffic conditions.

Using the algorithm proposed in this paper, we only need to set a group of initial thresholds for $Final\_F$ firstly, and then the thresholds are dynamically adjusted according to the actual traffic condition. Compared with the adaptive threshold vehicle detection algorithms mentioned in [7] and [16], though the thresholds are all adjusted dynamically, the signal used to detect vehicles and the adjustment process of the thresholds are different. $Final\_F$ used in the paper can reduce the influences caused by the signal drift and the vehicles on the adjacent road. The adjustment of the thresholds in the paper does not merely depend on the history signal, which guarantees the vehicle detection result less affected by the front vehicles. Our proposed algorithm can apply to different traffic conditions and fit the urban traffic.

IV. Speed Estimation

To estimate the individual vehicle speed of passing vehicles, two roadside and longitudinally deployed sensor nodes are used in this sensing system. The speed of passing vehicle can be given by

$$V_i = \frac{S}{\Delta T}, \tag{7}$$

where S is the distance between two sensors and ΔT is the time interval of a vehicle passing by them. To measure the time interval more precisely, the cross correlation between the two sensor signals along with the maximum values is used in this paper. Note that the sensor signal used here is $Filtered\_F$.

As the two sensor nodes are close to each other, we can assume that the speed is nearly constant as the vehicle passing by the two sensors. The signals of sensor 1 and 2 are similar and we can calculate the cross correlation between them to

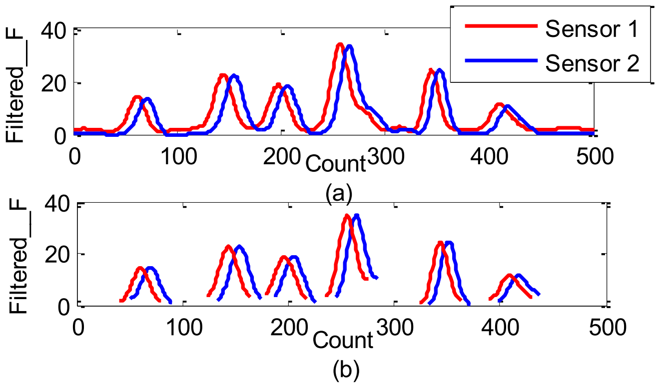

Fig. 15. (a) Signals of sensor 1 and 2. (b) The extracted vehicle signals based on the maximum values.

TABLE I

SIMILARITY BETWEEN B_1'' AND B_2''

	SUV	Two-box	Van	Saloon
				
Count	6	2	5	67
Average correlation coefficient	0.9838	0.9898	0.9866	0.9813

estimate the vehicle speed [10], [11]. Figure 15(a) shows the two signals when there are 6 vehicles passing by the two sensors in about 20 seconds. In urban traffic, vehicles always pass by the magnetic sensor in succession. To estimate the speed of each vehicle, extracting the effective part of each vehicle signal is important. As the conventional vehicle speed estimation method seldom describes how to extract vehicle signal in detail, we take the signal extraction method used in vehicle classification as a reference. It is always based on the detection points [17] and the vehicle signals of sensor 1 and 2 can be represented as

$$\begin{cases} B_1 = \{F_1(m), F_1(m+1), F_1(m+2), \ldots, F_1(m+k1)\} \\ B_2 = \{F_2(n), F_2(n+1), F_2(n+2), \ldots, F_2(n+k2)\}, \end{cases} \tag{8}$$

where F_1 and F_2 are the signals of sensor 1 and sensor 2, respectively. $F_1(m)$ and $F_2(n)$ are just bigger than the high thresholds, $F_1(m+k1)$ and $F_2(n+k2)$ are just smaller than the low thresholds. However, one drawback of this method is that the vehicle signal extraction is greatly affected by the high and low thresholds. If the high threshold is set slightly lower, the vehicle entering points will be detected earlier. Finding the right time vehicles entering and leaving the sensor detection zone are difficult. The speed estimation result will be heavily dependent on the thresholds, especially in low sampling frequency conditions.

We extract the effective parts of the vehicle signals based on their maximum values B_1 and B_2, and the maximum values of sampling points are defined as $n_{1\max}$ and $n_{2\max}$, respectively. Now we can use N points around $n_{1\max}$ and $n_{2\max}$ to replace the whole vehicle signals of sensor 1 and 2. The signals can be represented as

$$\begin{cases} B_1' = \{F_1(n_{1\max}-(N-1)/2), \ldots, F_1(n_{1\max}), \ldots, \\ \qquad F_1(n_{1\max}+(N-1)/2)\} \\ B_2' = \{F_2(n_{2\max}-(N-1)/2), \ldots, F_2(n_{2\max}), \ldots, \\ \qquad F_2(n_{2\max}+(N-1)/2)\}. \end{cases} \tag{9}$$

To ensure that more accurate matching points can be selected, N is experimentally set in this paper to 51, approximately 2 times the signal sampling frequency. Figure 15(b) shows the schematic view of signal segmentation of the above six vehicles. As a result, our proposed algorithm takes the maximum values of the sensor signals into account besides the cross correlation of them, and thus can extract effective parts from the sensor signals.

With the low sampling frequency in our system, the sampling points are limited. We know that the sampling frequency is f_s, so the maximum error of the time interval can reach up to $1/f_s$ seconds. The curve fit model method with the cubic equation is used to fit the experiment data to improve the accuracy of speed estimation. Then a number of equally spaced sampling points are used to represent the result and the oversampling frequency is set to $10f_s$, with the consideration of calculation and accuracy. The two oversampled and curve fitted sensor signals of a vehicle can be represented as B_1'' and B_2'', and the length of them are both $10N$. Then, B_2'' can be time-shifted to the left or right to find the number of curve fitted samples required that produces maximum cross-correlation between B_1'' and the time-shifted version of B_2''.

To verify the similarity between B_1'' and B_2'', we tested 80 vehicles passing by the sensors in succession, which contains 6 SUVs, 2 two-box cars, 5 vans and 67 saloons. The correlation coefficient is used as criterion and Table I shows the results. The average correlation coefficient is higher than 0.98 for each type of the vehicles. As a result, B_1'' and B_2'' are similar to each other.

We define n_d as the number of sample delay.

$$n_d = n_{1\max} - n_{2\max} + \frac{\arg\max_m r(m)}{10}, \tag{10}$$

where

$$r(m) = \sum_{n=1}^{10N} B_1''(n) B_2''(n-m), \quad -(10N-1) \le m \le 10N-1. \tag{11}$$

The Fast Fourier Transform (FFT) is used to compute $r(m)$ to reduce the amount of computation required. The method is described as follows.

Firstly we find a number called P, which meets

$$2^{P-1} \le 10N + (10N-1) \le 2^P. \tag{12}$$

Then fill B_1'' and B_2'' with zeros to a length of 2^P and we obtain

$$\tilde{B}_1''(n) = \begin{cases} B_1''(n) & (n = 1, 2, \ldots, 10N) \\ 0 & (n = 10N + 1, \ldots, 2^P) \end{cases}$$
$$\tilde{B}_2''(n) = \begin{cases} B_2''(n) & (n = 1, 2, \ldots, 10N) \\ 0 & (n = 10N + 1, \ldots, 2^P). \end{cases} \quad (13)$$

Define

$$X_1(k) = FFT(\tilde{B}_1''(n))$$
$$X_2(k) = FFT(\tilde{B}_2''(n)) \quad (k, n = 1, 2, 3, \ldots, 10N). \quad (14)$$

Now $r(m)$ can be computed by

$$r(m) = IFFT(X_1(k) * \bar{X}_2(k)) \quad (k = 1, 2, 3, \ldots, 10N)$$
$$(m = 0, 1, 2, \ldots, 10N - 1), \quad (15)$$

where $\bar{X}_2(k)$ is the conjugation of $X_2(k)$. Exchanging the order of $X_1(k)$ and $X_2(k)$, then we get

$$r(m) = IFFT(X_2(k) * \bar{X}_1(k)) \quad (k = 1, 2, 3, \ldots, 10N)$$
$$(m = 0, -1, \ldots, -(10N - 1)). \quad (16)$$

Now that $r(m)$ has been determined, we can obtain n_d based on (10). The time delay of a vehicle passing by the two sensors can be calculated by

$$\Delta T = \frac{n_d}{f_s}. \quad (17)$$

Finally, the speed of passing vehicles can be computed by (7). We can extract the vehicle signals of sensor 1 and 2 with high similarity, and thus estimate the vehicle speed even the vehicles crawl bumper to bumper in the complex urban traffic.

V. Experimental Results and Analysis

A. Vehicle Detection

Several experiments were conducted on different roads in Haidian District, Beijing, and the tests were performed at different times under different traffic conditions, and we can observe the adaptability of the algorithm proposed in this paper. The test duration of each experiment is about 10 minutes and the vehicle detection results are listed in Table I. Note that the high and low thresholds used in our previous method are 60 and 50, respectively.

Table II shows the detection accuracy. After using the dynamic threshold multi-state machine algorithm, the detection accuracy can reach up to 100% with the low traffic volume. With the increasing of traffic flow, a slight reduction in detection accuracy is observed. This is probably caused by the increased complexity of the sensor signal on busy roads. The accuracy is still over 95% on road 3, highlighting the adaptability of the proposed approach under different traffic conditions. As a whole, the average accuracy can reach up to 97.92%, increased by 5.62% compared with the previous algorithm. The improved vehicle detection algorithm is adapted to changeable traffic flow and has significant improvement on accuracy.

TABLE II
Results of Vehicle Detection

Experiment	Observed Vehicles	Vehicle Detection Results		Correct (%)	
		Our Previous Method In [7]	Current Method	Our Previous Method In [7]	Current Method
Road 1	4	4	4	100	100
Road 2	63	60	63	95.24	100
Road 3					
1	147	138	144	93.88	97.96
2	168	151	162	89.88	96.43
3	174	156	166	89.66	95.40
4	185	166	182	89.73	98.38
5	152	136	148	89.47	97.37
Average	-	-	-	92.3	97.92

Road 1, 2, 3 represent three different roads. Road 1 is relatively quiet, and the speed limit is 8.3 m/s. Road 2 and road 3 are two auxiliary roads in Beijing, with the speed limitation of 13.9 m/s. In addition, road 3 has heavier traffic than road 2. Five experiments conducted on *road 3* were performed at different times from morning to afternoon, which correspond to different traffic conditions.

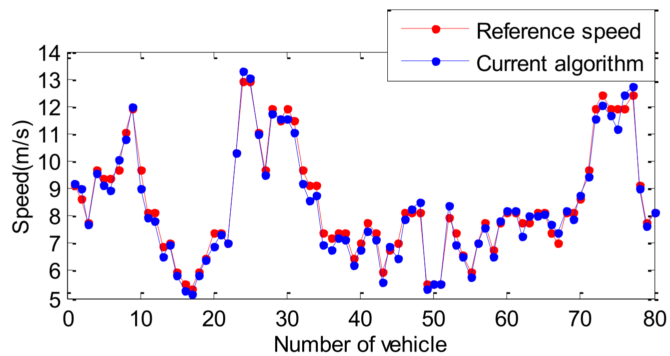

Fig. 16. Speed estimation results using our improved speed algorithm.

B. Speed Estimation

To verify the speed estimation accuracy, we tested 80 vehicles which passed by the sensors sequentially on Xueyuan Road in Haidian District, Beijing. A video was used as a reference to measure the time delay of a vehicle passing by the two reference lines. The distance of the two lines is 8m and the resolution of time is 1/50 second. Figure 16 shows the speed estimation accuracy of the 80 vehicles.

For the above 80 vehicles, the average speed estimation accuracy can reach up to 97.11%. During the test, a few vehicles passed by the sensors with non-uniform speed, which results in the poor similarity between the two sensor signals, and an increase in the error of speed estimation

Although the speed estimation result is still influenced by the actual vehicle movement, the algorithm proposed in this paper can reach good accuracy as a whole and it is more stable and adaptable.

VI. Conclusion

This paper focuses on the urban traffic, and presents an adaptable roadside vehicle detection and speed estimation system with triaxial anisotropic magnetoresistive sensors.

A novel dynamic threshold multi-state machine algorithm is proposed, which takes the actual changeable traffic condition into account. The speed estimation is based on the maximum values of the two sensor signals of passing vehicles and the cross-correlation between them. The experimental results show that the proposed algorithms have significant improvement in accuracy and adaptability, which verifies that this sensor system is promising for application in urban traffic.

In this paper, we concern about the estimation of moving vehicles, and will investigate vehicle detection for severe congested traffic, i.e., the stalled traffic, in the near future.

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